Quantification: 701_z36c1-2
==========================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	19
interior lacuna count	11
frame-edge lacuna count	8

Interior lacunae only: mean, median and sample SD (n = 11)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1504.36	897	1753.25	px^2
major_axis_length_px	lacuna shape	65.8954	54.1218	44.4863	px
minor_axis_length_px	lacuna shape	32.4977	24.7285	15.3599	px
aspect_ratio	lacuna shape	2.0209	1.7172	0.7947	unitless
eccentricity	lacuna shape	0.8034	0.8129	0.1253	unitless
solidity	lacuna shape	0.7542	0.7412	0.1221	unitless
roots_count	headline	10.0909	9	4.5267	count
ring_length_r30_px	headline	452.818	399	174.489	px
ring_length_r60_px	reported	1060.45	917	389.391	px
owned_length_px	ownership-dependent	1931.93	1276.12	1456.71	px
edge_count	ownership-dependent	140.091	86	108.58	count
mean_edge_length_px	ownership-dependent	14.3631	13.9313	1.5558	px
perimeter_px	normalised	235.331	179.622	166.75	px
ring_area_r30_px2	normalised	7434.18	6781	3418.26	px^2
ring_area_r60_px2	normalised	17465.8	16160	7147.61	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.994	1	0.0198	unitless
ring_density_r30	normalised	0.0640281	0.0553758	0.0191072	px^-1
ring_density_r60	normalised	0.0629206	0.0574034	0.0151312	px^-1
roots_per_100px_perimeter	normalised	5.146	3.7911	2.4252	per 100 px
ring_attached_length_r30_px	network v2	373.182	331	166.606	px
ring_attached_length_r60_px	network v2	733.182	593	374.894	px
ring_length_w_r30_px	network v2	512.676	461.031	197.925	px
ring_length_w_r60_px	network v2	1197.82	1035.59	441.756	px
sholl_crossings_r10	network v2	9.5455	10	3.5879	count
sholl_crossings_r20	network v2	11.4545	10	4.3672	count
sholl_crossings_r30	network v2	13.4545	14	4.3672	count
loop_count	loops, report only	12.3636	5	16.6929	count
loop_edge_share	loops, report only	0.269	0.2595	0.1732	share
ring_width_mean_r30_px	width	6.0698	6.2087	1.3318	px

Per field
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measure	value	units
analysed_area_px2	718014	px^2
total_skeleton_length_px	45074	px
canalicular_length_density_per_px	0.0627759	px^-1
junction_count	1477	count
junction_density_per_px2	0.00205706	px^-2
lacuna_count	19	count

lacunae_per_px2	2.646e-05	px^-2
skeleton_component_count	621	count
mean_component_length_px	72.5829	px
median_component_length_px	12	px
field_length_density_w_per_px	0.0707161	px^-1
field_density_without_flagged_per_px	0.0441332	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	92	px
unattached_component_count	7	count
canalicular_length_density_attached_per_px	0.0627759	px^-1
field_length_density_with_unattached_per_px	0.0629041	px^-1
excluded_area_px2	278639	px^2
analysed_field_area_px2	769937	px^2
loop_count	325	count
loop_edge_share	0.2852	share
edge_count	4610	count

Canalicular width, the new measure

measure	value	units
width_mean_px	6.2674	px
width_median_px	6	px
width_p10_px	2.8284	px
width_p90_px	10	px
width_pixels_used	38285	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	4432	103.113	57.3862	1.7968	0.8308	0.8752	0.7863	35.4294	265.904	True
2	5447	154.686	50.1635	3.0836	0.946	0.7894	0.5665	62.9728	466.817	True
3	7233	130.507	74.6116	1.7492	0.8205	0.8879	1.2582	34.0104	969.455	True
4	1625	83.1546	31.6031	2.6312	0.925	0.7835	0.114	172.29	1011.39	True
5	804	41.5213	33.4623	1.2408	0.592	0.6949	-0.5169	176.293	670.124	False
6	437	31.4519	18.3155	1.7172	0.8129	0.8918	1.4887	285.545	929.915	False
7	843	61.4626	24.6873	2.4896	0.9158	0.7274	1.4958	308.81	200.344	False
8	1362	78.3413	37.8814	2.0681	0.8753	0.558	0.4269	332.607	242.698	False
9	1410	54.1218	34.645	1.5622	0.7683	0.921	-1.5551	413.697	828.063	False
10	9556	142.266	91.5717	1.5536	0.7653	0.9091	1.4115	461.649	963.906	True
11	481	29.6358	22.2683	1.3309	0.6599	0.8698	1.5129	580.47	966.104	False
12	6510	186.09	58.4873	3.1817	0.9493	0.6768	-1.566	684.663	733.562	False
13	2377	86.3084	62.7585	1.3752	0.6865	0.5865	-0.3235	741.99	862.039	False
14	994	72.7028	20.0045	3.6343	0.9614	0.7412	1.4325	768.803	413.569	False
15	1166	44.1476	35.3039	1.2505	0.6004	0.9276	0.2819	791.344	1008.62	True
16	897	53.6979	24.7285	2.1715	0.8877	0.766	-1.5247	884.094	479.08	False
17	433	29.5156	20.2362	1.4586	0.728	0.8625	0.4576	943.721	446.413	False
18	3880	120.945	42.3369	2.8567	0.9367	0.9282	-1.4866	1004.7	384.256	True
19	2460	98.2976	35.4294	2.7745	0.9328	0.9004	1.5163	1009.82	777.268	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	10	453	1162	2764.7	194	14.251	6.5032
2	14	683	1606	2690.02	184	14.6196	6.5137
3	11	429	835	1819.53	145	12.5485	6.7812
4	9	333	676	647.115	49	13.2064	6.1216
5	8	443	1213	1932.6	158	12.2317	5.8682
6	7	291	894	1276.12	86	14.8386	7.4536
7	6	225	495	289.522	16	18.0951	4.8623
8	10	399	905	734.512	53	13.8587	6.9731
9	6	365	917	816.541	52	15.7027	4.7742
10	9	454	892	911.524	71	12.8384	6.4637
11	4	363	825	1085.05	74	14.6628	6.2087
12	17	766	1721	2006.11	144	13.9313	4.6308
13	14	439	1150	4008.34	274	14.629	4.0253
14	17	714	1652	4055.69	306	13.2539	6.9086
15	6	284	736	846.968	61	13.8847	6.0351
16	13	627	1282	4131.21	310	13.3265	6.9924
17	9	349	611	915.553	68	13.464	8.0706
18	6	489	1002	5679.42	438	12.9667	7.2113
19	4	260	778	207.267	12	17.2723	

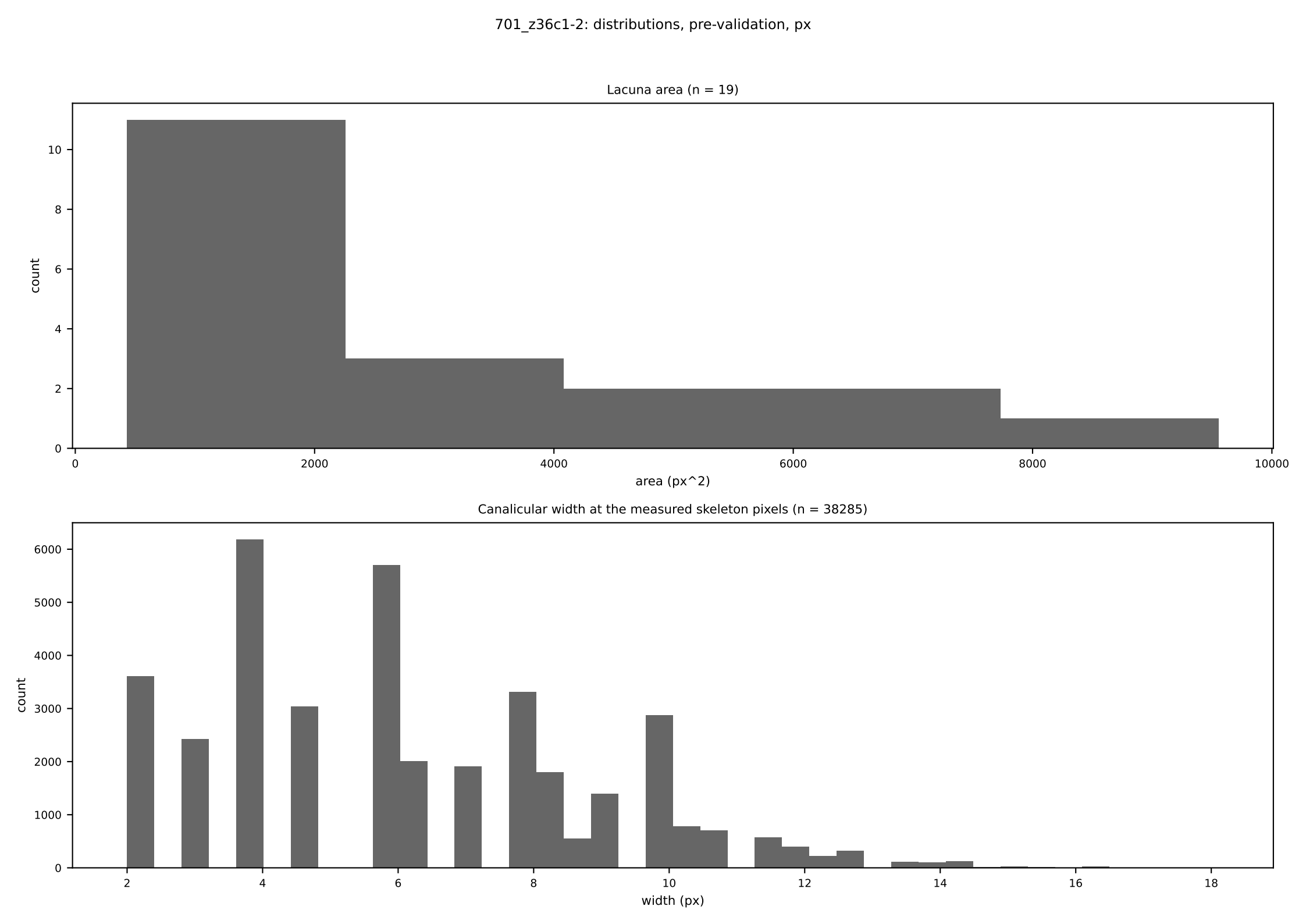
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	375.375	8196	18822	0.7168	0.6687	0.0552709	0.0617363	2.664
2	544.968	12171	26429	0.8379	0.7759	0.056117	0.0607666	2.569
3	492.416	6624	12045	0.4791	0.4258	0.0647645	0.0693234	2.2339
4	303.842	5339	10334	0.5936	0.5634	0.0623712	0.0654151	2.9621
5	227.635	6849	19116	1	1	0.064681	0.0634547	3.5144
6	94.468	5255	15574	1	1	0.0553758	0.0574034	7.4099
7	173.794	4559	10794	1	1	0.0493529	0.0458588	3.4524
8	267.019	7579	18429	1	1	0.0526455	0.0491074	3.745
9	158.267	6781	16160	1	1	0.0538269	0.056745	3.7911
10	551.263	10755	18353	0.7333	0.694	0.0422129	0.0486024	1.6326
11	101.225	5450	12434	1	0.9344	0.0666055	0.0663503	3.9516
12	628.95	16115	34097	1	1	0.0475334	0.0504737	2.7029
13	461.47	10803	23090	1	1	0.0406369	0.0498051	3.0338
14	198.309	7886	20602	1	1	0.0905402	0.0801864	8.5725
15	148.125	4251	11046	0.6223	0.5852	0.0668078	0.0666304	4.0506
16	179.622	6400	14669	1	1	0.0979687	0.0873952	7.2374
17	97.8823	4099	7159	1	1	0.0851427	0.0853471	9.1947
18	315.907	6243	12280	0.6169	0.5909	0.0783277	0.0815961	1.8993
19	302.007	5690	14075	0.5535	0.5397	0.0456942	0.0552753	1.3245

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	392	950	507.233	1321.4	12	12	11
2	597	1129	769.571	1831.75	16	14	18
3	421	700	487.019	945.11	12	8	9
4	320	493	371.664	758.955	9	7	7
5	372	884	508.617	1375.79	11	9	18
6	274	593	338.22	1003.15	7	9	9
7	128	391	250.965	560.458	3	9	6
8	331	424	461.031	1035.59	9	10	9
9	214	356	409.534	1031.16	10	7	14

10	388	544	513.304	993.08	11	14	11
11	302	606	403.149	923.382	9	10	14
12	574	1108	857.453	1935	16	22	18
13	319	478	483.576	1280.99	11	12	14
14	640	1501	821.955	1895.63	12	17	18
15	234	539	323.25	826.169	6	7	8
16	617	1148	708.931	1438.21	12	12	18
17	334	576	396.007	696.63	5	9	10
18	475	843	541.019	1118.6	12	14	10
19	176	405	282.966	878.838	7	10	13


701_z36c1-2: distributions, pre-validation, px
Lacuna area (n = 19)
count
area (px^2)
0
2
4
6
8
10
0
2000
4000
6000
8000
10000
Canalicular width at the measured skeleton pixels (n = 38285)
count
width (px)
0
1000
2000
3000
4000
5000
6000
2
4
6
8
10
12
14
16
18